QUASI-COHERENCE AND AFFINE COMPUTATION OF THE DERIVED INTERNAL HOM OUT OF PSEUDO-COHERENT OR PERFECT OBJECTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The internal hom construction in the derived category of O-modules commutes with localization. invoked at proof lines 2687-2689. Reduces parts (1) and (2) to affine opens of X.
- On affines the derived quasi-coherent categories compare with module categories, pseudo-coherent and perfect objects correspond to bounded above and finite complexes of finite projective modules. invoked at proof lines 2690-2693. Reduces (1) and (2) to the affine statement (3).
- The internal hom of complexes represented by finite projective complexes is computed termwise, by the cohomology Rhom lemma for complexes of direct summands of finite free modules. invoked at proof lines 2695-2699. Proves part (3) in both the pseudo-coherent and the perfect case.
- On a ringed space the nth cohomology sheaf of the internal hom is the sheaf associated to the presheaf of Ext groups on opens. invoked at proof lines 2702-2708. Gives the presheaf formula of part (4) on principal opens.

Source reference. Original section 08DH: sources/stacks-chow/tags.txt, source lines 10637–10637. Cited at proof lines 2688-2689 and 2707-2708 for the localization and presheaf descriptions; the reductions are inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 2690-2693 and 2714-2715 for the affine comparison; the reductions are inside the counted span.

Source reference. Original lemma 08E7: sources/stacks-perfect/pinned-original-source.tex, source lines 2440–2440. Cited at proof lines 2691-2692 to reduce (1) to (3); the module computation is inside the counted span.

Source reference. Original lemma 08EB: sources/stacks-perfect/pinned-original-source.tex, source lines 2636–2636. Cited at proof lines 2691-2693 to reduce (2) to (3); the module computation is inside the counted span.

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Source reference. Original lemma 08I5: sources/stacks-chow/tags.txt, source lines 10803–10803. Cited at proof lines 2696-2698 to prove part (3); the representation of K and L is inside the counted span.

Lemma 0.2. *Let X be a scheme.*

(1) *If L is in $D^+_{QCoh}(\mathcal{O}_X)$ and K in $D(\mathcal{O}_X)$ is pseudo-coherent, then $R\mathcal{H}om(K, L)$ is in $D_{QCoh}(\mathcal{O}_X)$ and locally bounded below.*

(2) *If L is in $D_{QCoh}(\mathcal{O}_X)$ and K in $D(\mathcal{O}_X)$ is perfect, then $R\mathcal{H}om(K, L)$ is in $D_{QCoh}(\mathcal{O}_X)$.*

(3) *If $X = \mathrm{Spec}(A)$ is affine and $K, L \in D(A)$ then*

$$R\mathcal{H}om(\widetilde{K}, \widetilde{L}) = R\widetilde{\mathrm{Hom}_A(K, L)}$$

in the following two cases
   (a) *K is pseudo-coherent and L is bounded below,*
   (b) *K is perfect and L arbitrary.*

(4) *If $X = \mathrm{Spec}(A)$ and K, L are in $D(A)$, then the nth cohomology sheaf of $R\mathcal{H}om(\widetilde{K}, \widetilde{L})$ is the sheaf associated to the presheaf*

$$X \supset D(f) \longmapsto \mathrm{Ext}^n_{A_f}(K \otimes_A A_f, L \otimes_A A_f)$$

for $f \in A$.

Proof. The construction of the internal hom in the derived category of $\mathcal{O}_X$ commutes with localization (see Cohomology, Section 08DH). Hence to prove (1) and (2) we may replace X by an affine open. By Lemmas 06Z0, 08E7, and 08EB in order to prove (1) and (2) it suffices to prove (3).

Part (3) follows from the computation of the internal hom of Cohomology, Lemma 08I5 by representing K by a bounded above (resp. finite) complex of finite projective A-modules and L by a bounded below (resp. arbitrary) complex of A-modules.

To prove (4) recall that on any ringed space the nth cohomology sheaf of $R\mathcal{H}om(A, B)$ is the sheaf associated to the presheaf

$$U \mapsto \mathrm{Hom}_{D(U)}(A|_U, B|_U[n]) = \mathrm{Ext}^n_{D(\mathcal{O}_U)}(A|_U, B|_U)$$

See Cohomology, Section 08DH. On the other hand, the restriction of $\widetilde{K}$ to a principal open $D(f)$ is the image of $K \otimes_A A_f$ and similarly for L. Hence (4) follows from the equivalence of categories of Lemma 06Z0. □